THE COZY SCIENCE LAB · ORIGINAL STUDY SERIES

Physics

A Grade 7–10 study handbook with concept notes, examples, self-check explanations and labelled illustrations.

In this handbook

- Motion & Projectiles
- Newton’s Laws & Forces
- Work, Energy & Heat
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- Electricity & Magnetism

Edition 1 · Lab-authored learning materials. Supporting publications are credited with each topic.
Examples are constructed for teaching; diagrams state their model and scale limits. For current Philippine hazard or weather decisions, consult the responsible agencies rather than a static study guide.

Motion & Projectiles

Describe motion with a reference frame, a measured time interval and clear directions.

Distance and displacement

Distance is the length of a path. Displacement describes the change from initial to final position, including direction. A round trip can have a nonzero distance and zero displacement.

Speed, velocity and acceleration

Average speed is total distance divided by elapsed time. Average velocity uses displacement instead. Acceleration is a change in velocity per time, so a change of direction can be an acceleration even at constant speed.

Projectiles as a model

Ignoring air resistance near Earth's surface, a projectile has downward gravitational acceleration. Horizontal and vertical motion can be analysed separately in this model.

Key terms

Scalar

A quantity described by magnitude, such as distance.

Vector

A quantity with magnitude and direction, such as displacement.

A round trip

A learner walks 30 m east and 30 m west in 60 s. Distance is 60 m and average speed is 1 m/s. Displacement and average velocity are zero.

Remember: Ignoring air resistance, a heavier object does not fall faster simply because it has greater mass.

Check your thinking

Can a moving object have zero average velocity for a trip?

Explanation: Yes. If it returns to its starting position, its displacement is zero even though it travelled a distance.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

OpenStax · Rice University. Physics. https://openstax.org/details/books/physics

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Newton’s Laws & Forces

Forces are interactions. Draw the forces on one chosen object before deciding how its motion changes.

Net force and inertia

An object keeps a constant velocity when the net force is zero in an inertial reference frame. A net force changes velocity; in a simple constant-mass model, net force = mass × acceleration.

Interaction pairs

Newton's third-law forces are equal and opposite forces acting on different objects. They do not cancel when calculating the net force on just one of those objects.

Mass and weight

Mass is measured in kilograms. Weight is a gravitational force measured in newtons. Near Earth's surface, weight is approximately mass × 9.8 m/s²; mass stays the same when local gravitational strength changes.

Key terms

Net force

The vector sum of the forces acting on an object.

Inertia

The tendency to retain a state of motion unless acted on by a net force.

A 3 kg cart

A net force of 12 N on a 3 kg cart produces an acceleration of 12/3 = 4 m/s² in the direction of the net force.

Remember: An object moving steadily does not require a forward net force. Resistive forces may need balancing, but balanced forces mean zero acceleration.

Check your thinking

Why do action–reaction forces not cancel the acceleration of one cart?

Explanation: The two forces act on different objects. Only forces acting on the chosen cart enter its net-force sum.

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Work, Energy & Heat

Energy transfers and transformations connect movement, work and thermal processes.

Work and power

For a constant force parallel to a displacement, work = force × distance. Work and energy are measured in joules. Power is energy transferred or work done per time and is measured in watts.

Mechanical energy

Kinetic energy is ½mv² in the ordinary low-speed model. Near Earth's surface, gravitational potential-energy change is approximately mg times the height change. Specify a system and reference level when tracking energy.

Heat and temperature

Heat is energy transfer caused by a temperature difference. Temperature describes a thermal condition, not the total energy stored in a body. Conduction, convection and radiation are mechanisms of energy transfer.

Key terms

Joule

The SI unit of energy, symbol J.

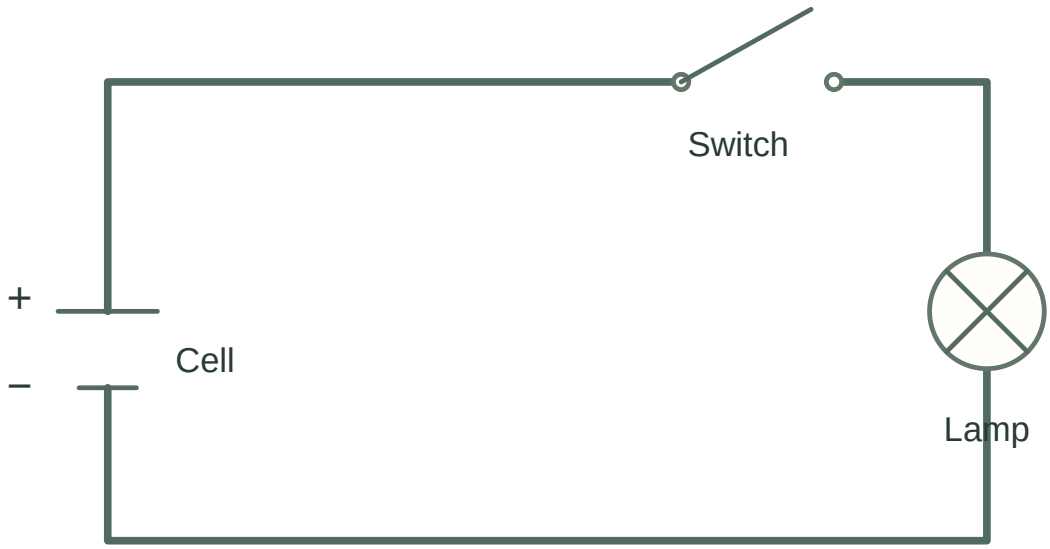
Watt

One joule transferred per second, symbol W.

Lifting a book

Lifting a 2 kg object upward by 5 m near Earth increases its gravitational potential energy by about 2 × 9.8 × 5 = 98 J.

Remember: Heat and temperature are different. A larger amount of a substance can have more internal energy at the same temperature as a smaller amount.



Close the path, light the lamp. Idealized circuit diagram. The cell transfers energy; charge is not used up by the lamp. Conventional-current arrows are shown only for a closed switch.

Check your thinking

If the same work is done in half the time, what happens to average power?

Explanation: Average power doubles because power is work divided by elapsed time.

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Waves & Sound

A wave transfers energy through a disturbance. Identify which direction the wave travels and how the medium, if present, moves.

Reading a wave

Wavelength is the spacing between corresponding points such as adjacent crests. Frequency counts cycles per second, in hertz. Amplitude measures maximum displacement from an equilibrium position in a displacement-wave model.

Wave relationships

Wave speed = frequency × wavelength. In a fixed medium with approximately constant speed, a higher frequency corresponds to a shorter wavelength. Period is the time per cycle and equals 1/frequency.

Sound needs matter

Sound travels as mechanical disturbances in a medium. In air, compressions and rarefactions form a longitudinal wave. Light is electromagnetic radiation and can travel through a vacuum.

Key terms

Hertz

One cycle per second, symbol Hz.

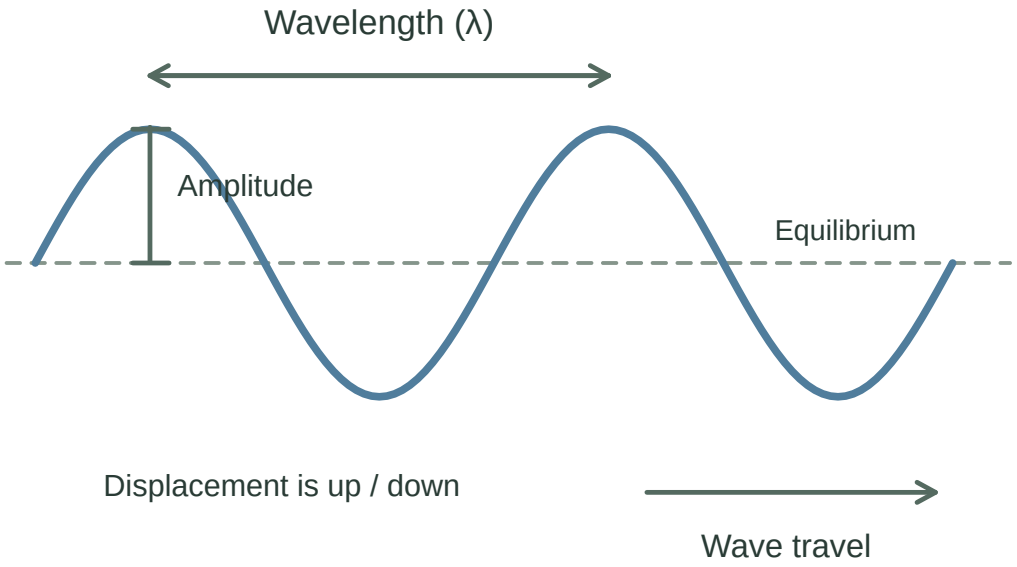
Equilibrium

The undisturbed reference position in a displacement model.

Finding wavelength

A wave travelling at 12 m/s with frequency 3 Hz has wavelength 12/3 = 4 m. Keep the units with the calculation.

Remember: Wave motion does not mean a piece of the medium travels with the wave all the way from source to receiver. Amplitude and frequency are different properties.



Read a transverse wave. A displacement-versus-position sketch, not a sound-pressure graph or a movie of particles travelling with a wave. Dimensions are illustrative.

Check your thinking

Why can astronauts use radio in space but not speak through the vacuum directly?

Explanation: Radio waves are electromagnetic. Ordinary sound needs a material medium to carry its disturbance.

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Light & the EM Spectrum

Visible light is one part of the electromagnetic spectrum. Models of waves and rays help explain familiar observations.

The spectrum

Radio, microwave, infrared, visible, ultraviolet, X-ray and gamma radiation are all electromagnetic. All travel at the same speed in a vacuum, approximately 3×10^8 m/s; their frequencies and wavelengths differ.

Reflection and refraction

Reflection angles are measured from the normal to a surface. Refraction can change a ray's direction when it enters a material in which its speed differs. Frequency remains the same across a stationary boundary.

Images and models

Mirrors and lenses redirect light. A converging lens can make different kinds of images depending on object distance. A ray diagram uses straight lines as a model of light paths in suitable conditions.

Key terms

Normal

A line perpendicular to a surface at the point considered.

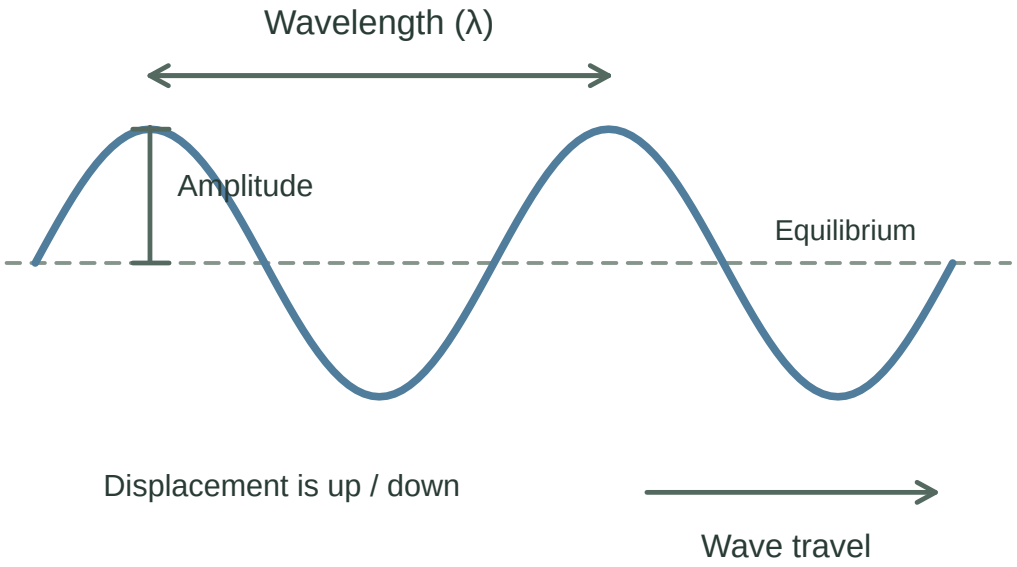
Refraction

A change in wave direction associated with a speed change across a boundary, except at normal incidence.

Frequency and wavelength

For electromagnetic waves in a vacuum, doubling frequency halves wavelength because their product remains the same speed.

Remember: Infrared is invisible electromagnetic radiation; it is not a separate material called heat. Light does not need air to travel.



Read a transverse wave. A displacement-versus-position sketch, not a sound-pressure graph or a movie of particles travelling with a wave. Dimensions are illustrative.

Check your thinking

Does blue light outrun red light in a vacuum?

Explanation: No. Both travel at the same vacuum speed. Their frequencies and wavelengths differ.

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Electricity & Magnetism

A circuit is a connected path for charge flow. Use labelled symbols and keep current, voltage and resistance distinct.

Three different quantities

Current is charge flow per time, in amperes. Potential difference, or voltage, is energy transferred per charge, in volts. Resistance describes the relation between voltage and current, in ohms.

Closed paths and branches

A steady current needs a complete conducting path. Series components share a path and carry the same steady current. Parallel branches connect across the same two nodes and have the same potential difference.

Electricity and magnetism

Electric current produces a magnetic field. Coils and suitable cores can strengthen electromagnets. Conventional current is defined in the direction positive charge would move; electrons in a metal drift the opposite way.

Key terms

Ampere

The SI unit of electric current, symbol A.

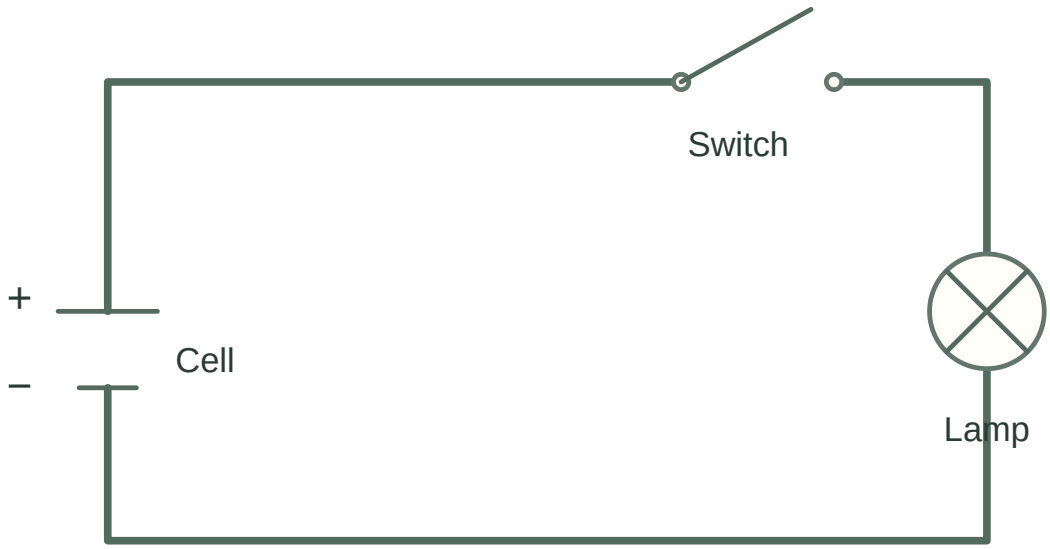
Ohm

The SI unit of resistance, symbol Ω.

Ohm's law

An ohmic resistor with resistance 3 Ω connected across 6 V has current I = V/R = 2 A, under conditions where that resistance stays constant.

Remember: A battery supplies energy to a circuit; it does not supply a fresh store of electrons that is used up by a lamp.



Close the path, light the lamp. Idealized circuit diagram. The cell transfers energy; charge is not used up by the lamp. Conventional-current arrows are shown only for a closed switch.

Check your thinking

Why does opening a series switch turn a lamp off?

Explanation: The break removes the complete conducting path required for a sustained current.

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